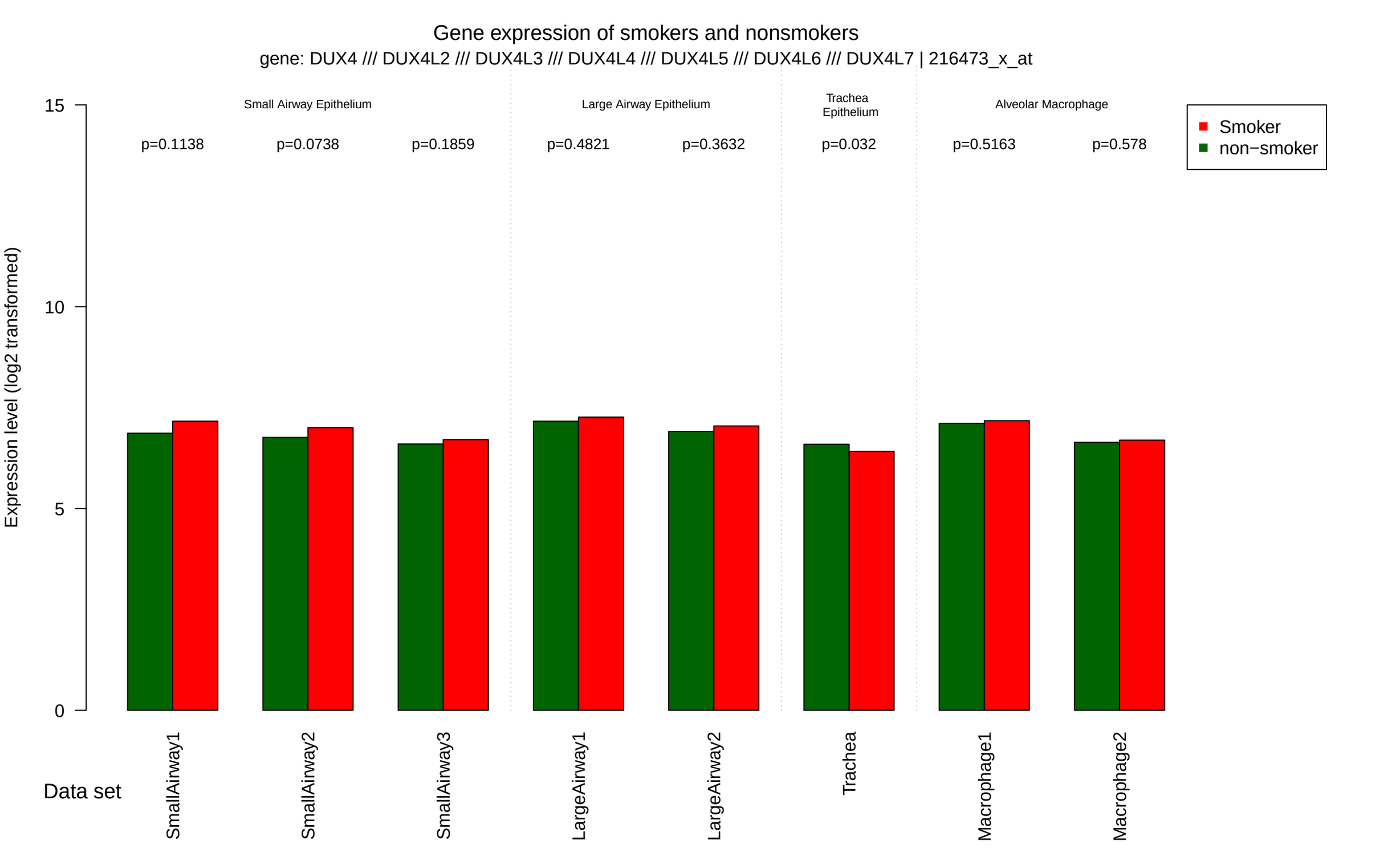
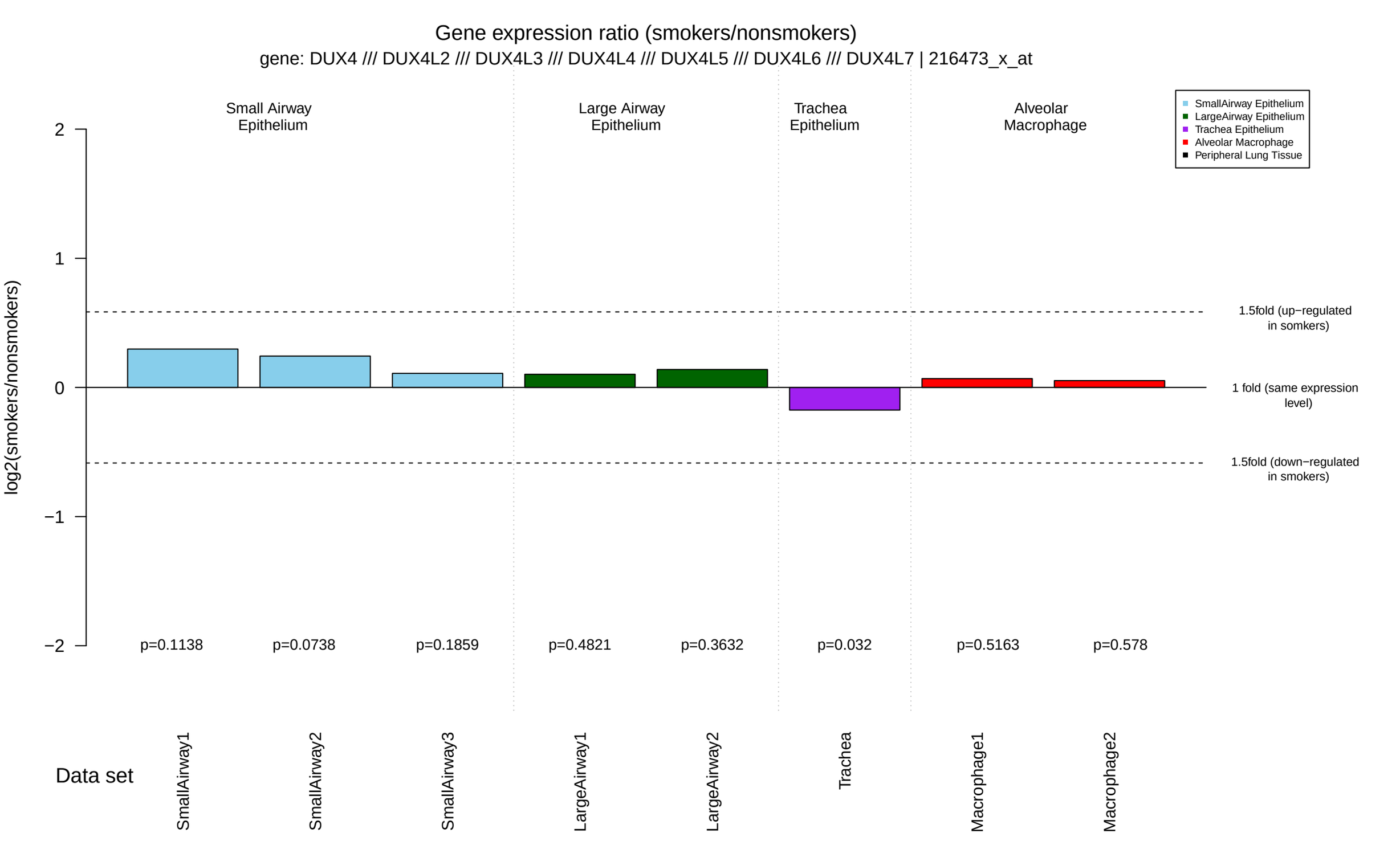
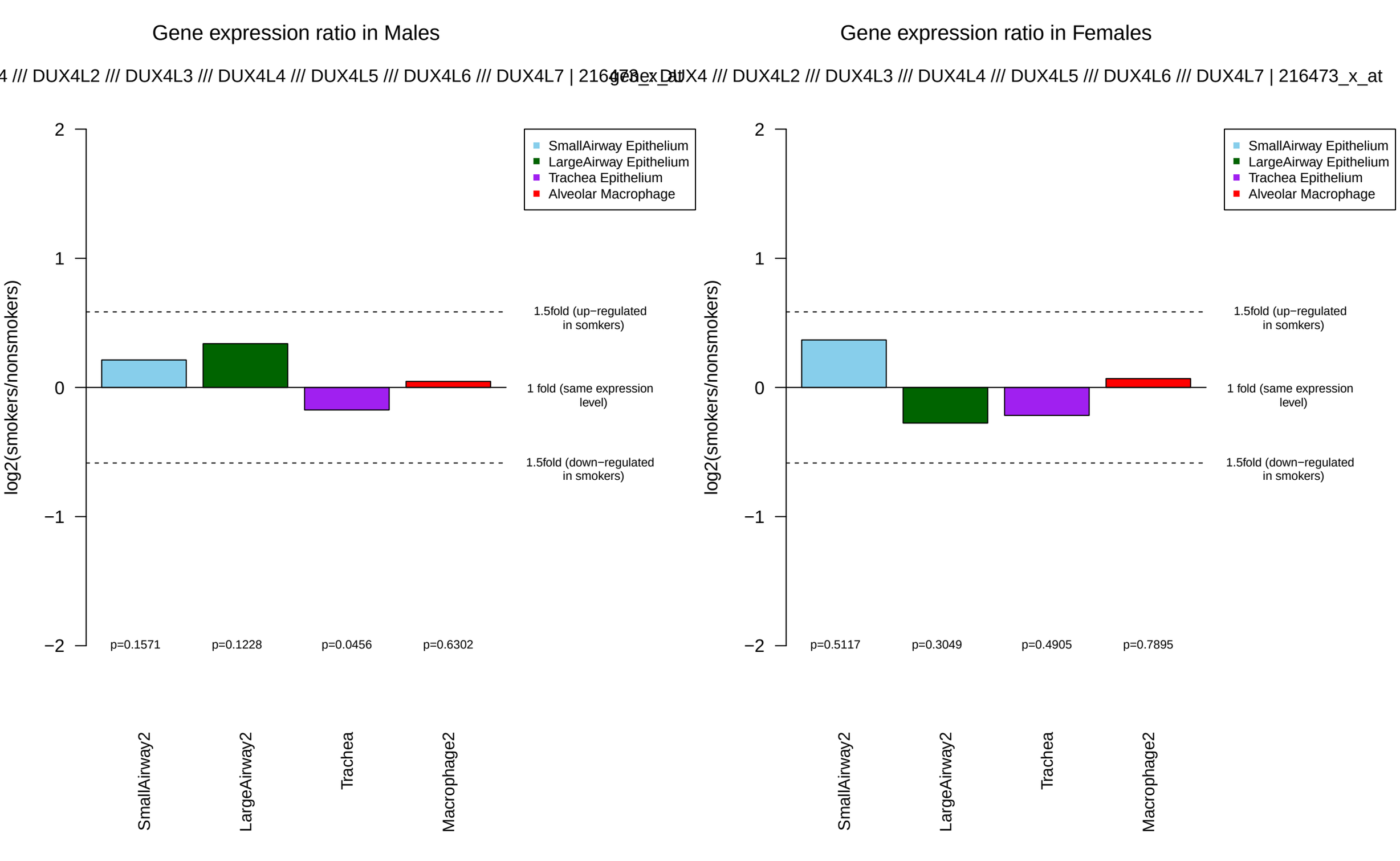
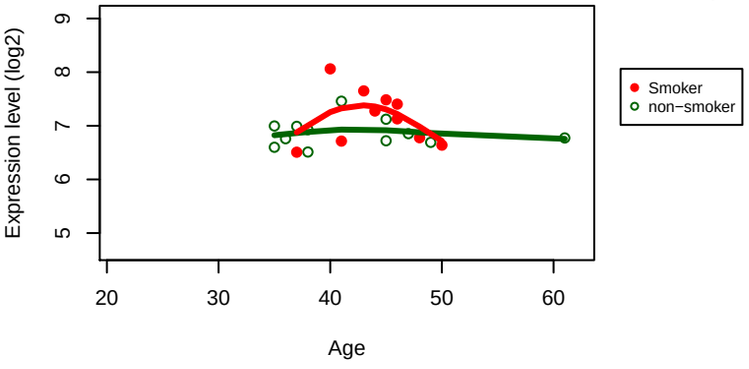

Gene expression of smokers and nonsmokers
gene: DUX4 /// DUX4L2 /// DUX4L3 /// DUX4L4 /// DUX4L5 /// DUX4L6 /// DUX4L7 | 216473_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1138
p=0.0738
p=0.1859
p=0.4821
p=0.3632
p=0.032
p=0.5163
p=0.578
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

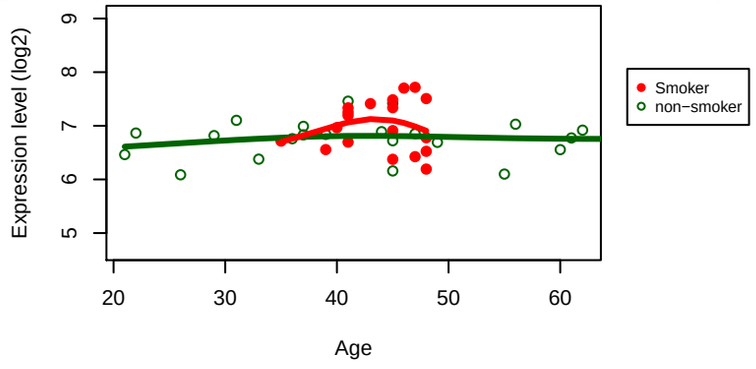

Gene expression ratio (smokers/nonsmokers)
gene: DUX4 /// DUX4L2 /// DUX4L3 /// DUX4L4 /// DUX4L5 /// DUX4L6 /// DUX4L7 | 216473_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1138
p=0.0738
p=0.1859
p=0.4821
p=0.3632
p=0.032
p=0.5163
p=0.578
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio in Males
4 /// DUX4L2 /// DUX4L3 /// DUX4L4 /// DUX4L5 /// DUX4L6 /// DUX4L7 | 216473_x_at
Gene expression ratio in Females
gene_at
DUX4 /// DUX4L2 /// DUX4L3 /// DUX4L4 /// DUX4L5 /// DUX4L6 /// DUX4L7 | 216473_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1571
p=0.1228
p=0.0456
p=0.6302
p=0.5117
p=0.3049
p=0.4905
p=0.7895
SmallAirway2
LargeAirway2
Trachea
Macrophage2

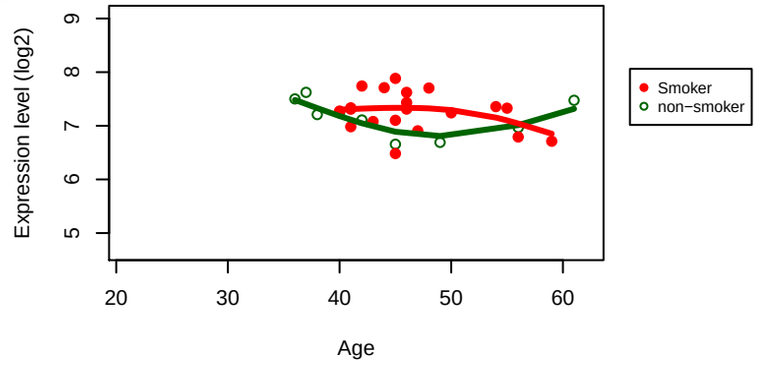
set: SmallAirway1 (p=0.1138)



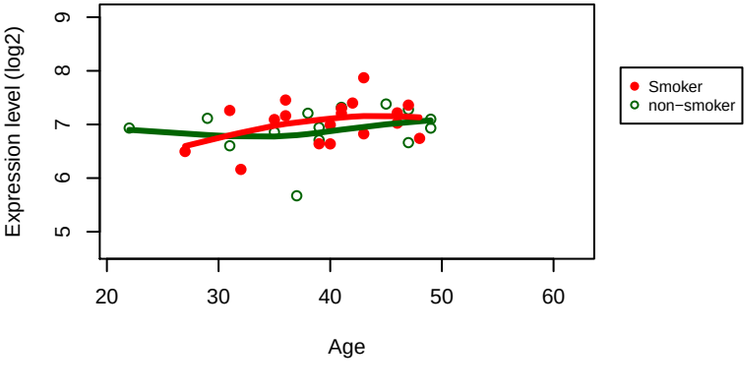
set: SmallAirway2 (p=0.0738)



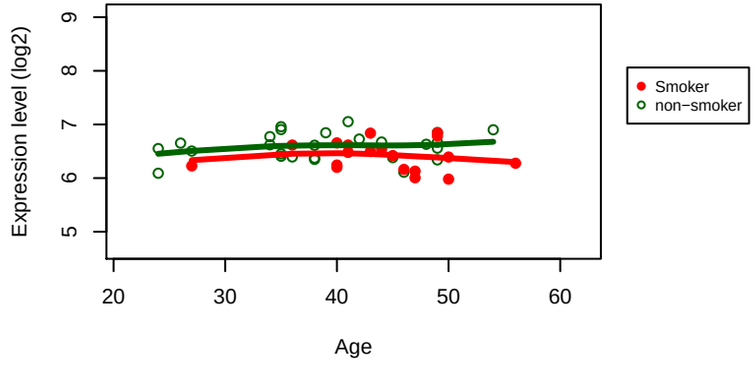
set: LargeAirway1 (p=0.4821)



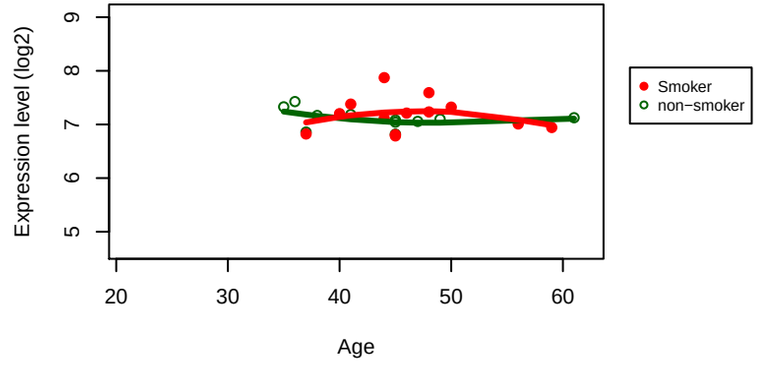
set: LargeAirway2 (p=0.3632)



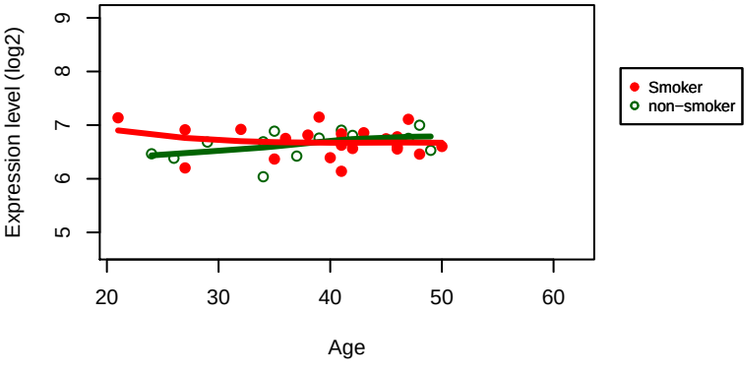
set: Trachea (p=0.032)

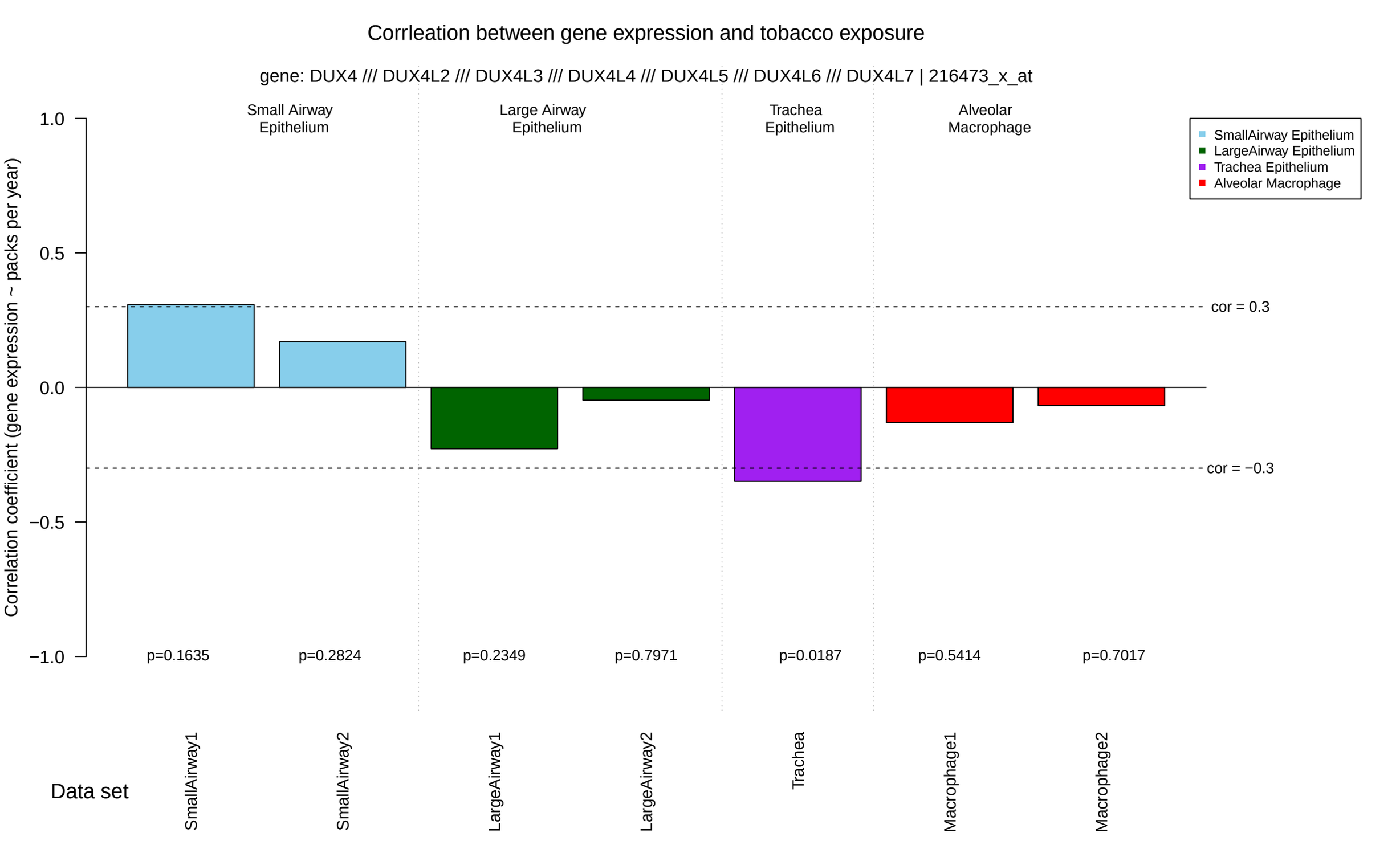


set: Macrophage1 (p=0.5163)



set: Macrophage2 (p=0.578)




Corrleation between gene expression and tobacco exposure
gene: DUX4 /// DUX4L2 /// DUX4L3 /// DUX4L4 /// DUX4L5 /// DUX4L6 /// DUX4L7 | 216473_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1635
p=0.2824
p=0.2349
p=0.7971
p=0.0187
p=0.5414
p=0.7017
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2